

Product Guide

SW-640L-TX-W & APO-DG2 / APO-DG2-PRO

Wireless Presentation and Conferencing System



WyreStorm

**Award-Winning Audio-Visual
Solutions Manufacturer**

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Product Overview



Powerful Wireless Presentation and Conferencing System

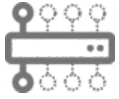
The “SW-640L-TX-W” wireless presentation switcher, when paired with the APO-DG2 or APO-DG2-PRO USB-C casting dongle, forms a powerful 4K wireless presentation and conferencing system. This setup ensures a seamless and superior hybrid meeting experience.

Built on the strengths of its predecessors, the **SW-540-TX-W** and **SW-220-TX-W** the **SW-640L-TX-W** offers 2x full-featured USB-C inputs and 2x HDMI inputs/outputs, providing unparalleled flexibility and options.

This system is ideal for small to medium-sized meeting rooms and classrooms, effortlessly meeting diverse practical needs and enhancing overall efficiency.

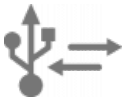


Main Features



Comprehensive Input Options

Offers a variety of inputs, including USB-C and HDMI, ensuring compatibility with devices used in conferences, meetings, and educational settings.



Full-Featured USB-C Support

The USB-C port provides power delivery, network, video, and USB transmission through a single cable, simplifying connections and maintaining a clean and tidy workspace.



Multiview Support & Dual HDMI Outputs

Displays up to 4 video sources simultaneously on one screen, ideal for data comparison or complex project collaboration. Also provides you flexible display options with dual HDMI outputs.



KVM Switch for Display

Includes USB 3.0 (Device and Host) ports for cameras, speakerphones, and keyboards. Perfect for hybrid meetings and education, it simplifies cable management and allows easy switching between devices using a single hub, streamlining workflows.



Centralized Control and Automation

Offers various control methods, such as RS232, Relay, API, and Web-UI, enabling remote monitoring and control of the presentation system. Provides real-time access to system status, configuration, and performance.



Reliable and High-Quality Resolution

4K resolution guarantees sharp and clear video, delivering a fully immersive presentation experience.



Balanced Audio De-embedding

Independent de-embedding audio works seamlessly with external audio systems. When paired with a professional amplifier, speaker, and mic, it creates a more immersive audio experience for presenters and participants.



Versatile and Flexible Ethernet

Advanced Ethernet capabilities provide a reliable network environment, enhancing interactive experiences, especially during hybrid meetings and education. Ethernet Bridge and Ethernet Isolation ensure a stable and secure usage experience.



Wireless Casting Support

Supports multiple presentation methods for classes or meetings, including Airplay mirroring, Miracast, and Dongle. (Chromecast support coming soon)

WyreStorm®

Powerful · Reliable · Simple

Why Choose Us: 1. One-Click Wireless Conferencing

Transform your meeting room with one-click wireless conferencing, ensuring easy connectivity, seamless integration, and enhanced productivity.

Effortless Connectivity

With just a single click of the button on the APO-DG2/APO-DG2-PRO USB-C dongle, you can control your USB conferencing devices wirelessly, eliminating the hassle of tangled cables.

Seamless Integration

Works flawlessly with cameras, speakerphones, video bars, touch-screen panels, and other USB 3.0 peripherals, delivering a seamless and superior hybrid meeting experience.

Boost Productivity

Quickly transition between meetings with minimal setup time, ensuring a smooth and productive workflow.

High-Quality Performance

Guarantees clear and reliable audio-visual communication, essential for effective collaboration and decision-making.

Why Choose Us: 2. Easy Sharing and Collaboration for Everyone

Make screen sharing, collaboration, and interaction easier for everyone, supporting various devices and operating systems to enhance interaction and engagement.

Multiple Casting Methods

Supports AirPlay, Miracast, USB-C dongle, and HDMI/USB-C wired connection, catering to various user preferences and devices.

Universal Compatibility

Works seamlessly with laptops, tablets, and smartphones, supporting major operating systems like Windows, macOS, and Android.

Interactive Learning

Perfect for classrooms, enabling teachers to quickly showcase teaching content while allowing students to interact using their own devices.

Engage Your Audience

Multiview support displays up to 4 sources simultaneously, making collaborative work and data comparison easier and more efficient.

Why Choose Us: 3. Plug & Play, Guest-Friendly

Simplify setup with plug-and-play functionality, making it easy for guests to connect and present without any hassle.

Simplified Setup

Plug-and-play functionality means guests can easily connect without the need for additional software or complex configurations.

Versatile Input Options

Featuring 2x full-featured USB-C ports (60W PD for guests' laptops) and 2x HDMI inputs, it provides unparalleled flexibility for various presentation needs.

Present in Seconds

Guests can simply pick up the dongle on the table and instantly share their screen, ensuring a swift and efficient presentation process.

Secure Network Access

Switch to "Isolated Mode" for network settings when you need to separate your internal network with guests' network. It ensures your guests can access network securely and stably, even while you are configuring the system.

Why Choose Us: 4. Effortless Device Management

Centralize and simplify device management to ensure a smooth and efficient operation for all your conferencing needs.

Centralized Control

Manage and monitor all connected devices through an intuitive web interface or all-in-one touchpad controller, simplifying the administration process.

Remote Configuration

Configure the system remotely via SYGMA cloud device management, ensuring that everything runs smoothly without needing to be physically present.

Flexible Network Settings

The presentation device is engineered with 3x Ethernet ports. Switch to "Isolated Mode" for enhanced security and network stability, ideal when you need to separate your internal network with guests' network. Or, go for "Transparent Mode" for hassle-free networking and flexible connections, perfect for efficient communication without restrictions.

More Advanced Control

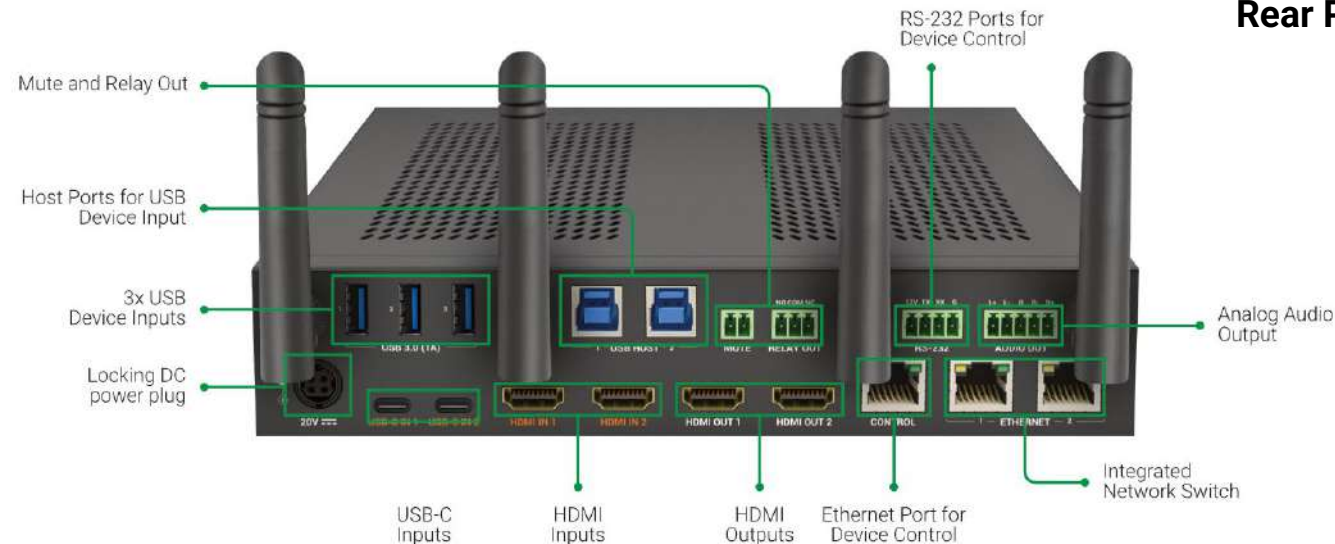
Utilize the "mute" port to connect to a PA sensor, allowing for quick muting of audio on HDMI OUT 1-2 and analog audio outputs during emergency broadcasts. Additionally, the "Relay Out" interface enables connection to a relay device for controlling functions such as raising or lowering the projection screen.

Interface of the Product

Front Panel



Rear Panel



Featuring:

3 Ethernet Ports

Ensures secure and stable network access for guests using their PCs, while allowing easy configuration of the system via a web interface.

Mute

Connect to the PA sensor to quickly mute audio on HDMI OUT 1-2 and analog audio out, in case of an emergency broadcast.

Relay Out

Connect to a relay device for relay control, such as raising or lowering the projection screen.

Specifications & Packing List

Main Specifications

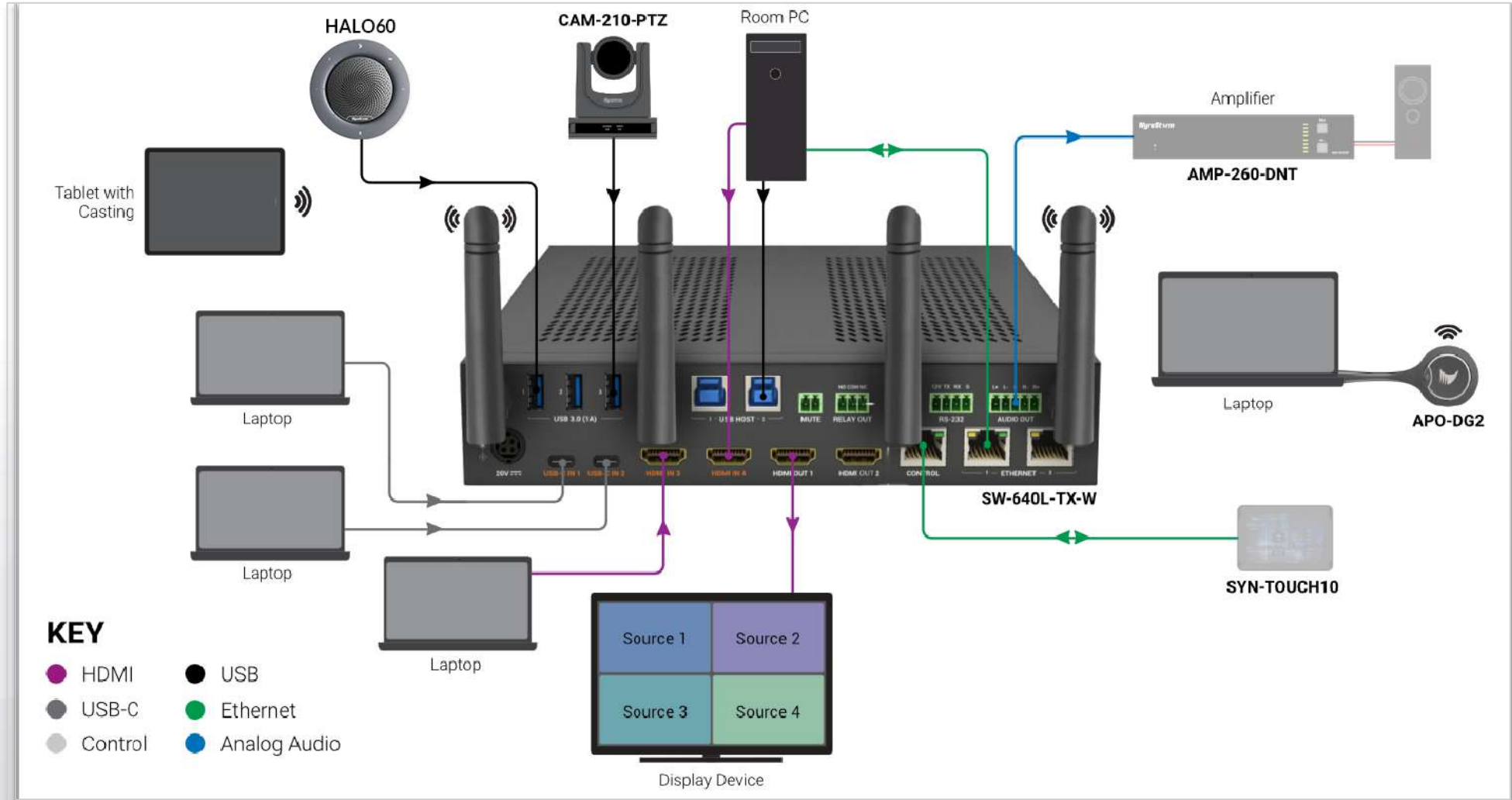
Input Video Port	2x USB-C, 2x HDMI
Output Video Port	2x HDMI
Output Video Signal	HDMI OUT 1: HDMI 2.0, HDCP 2.2 HDMI OUT 2: HDMI 1.4, HDCP 1.4
Input Resolutions (Maximum)	USB-C/HDMI: 3840x2160 (2160p30) Miracast (Wi-Fi): 1920x1080 (1080p60) Airplay Mirroring (Wi-Fi): Up to 1920x1080 (1080p60) Dongle (LAN/Wi-Fi): 3840x2160 (2160p30)
Output Resolutions (Maximum)	HDMI OUT 1:3840x2160 (2160p60) HDMI OUT 2:3840x2160 (2160p30)
Input Audio Port	2x HDMI, 2x USB-C, 3x LAN, 2x Wi-Fi
Output Audio Port	2 x HDMI, 1 x Analog Audio Out
Control Method	Front panel buttons, LAN (Web UI & Telnet API), RS-232
Power Supply	20V 10A DC

What's in the Box?

- ☒ 1x SW-640L-TX-W
- ☒ 1x 20V/10A Power Supply (US/UK/EU/AU)
- ☒ 4x Plastic Mounting Brackets
- ☒ 1x 2-pin Phoenix Connector (for PA Sensor)
- ☒ 1x 3-pin Phoenix Connector (for Relay)
- ☒ 1x 4-pin Phoenix Connector (for RS-232)
- ☒ 1x 5-pin Phoenix Connector (for Audio Out)
- ☒ 4x Wi-Fi Antennas
- ☒ 1x Quick Start Guide

Typical Connection & Applications

The SW-640L-TX-W system seamlessly integrates with all your conference devices, like room PCs, laptops, displays, webcams, speakerphones, and amplifiers, fulfilling every requirement for video presentations and remote conferencing. Please refer to the typical connection diagram below.



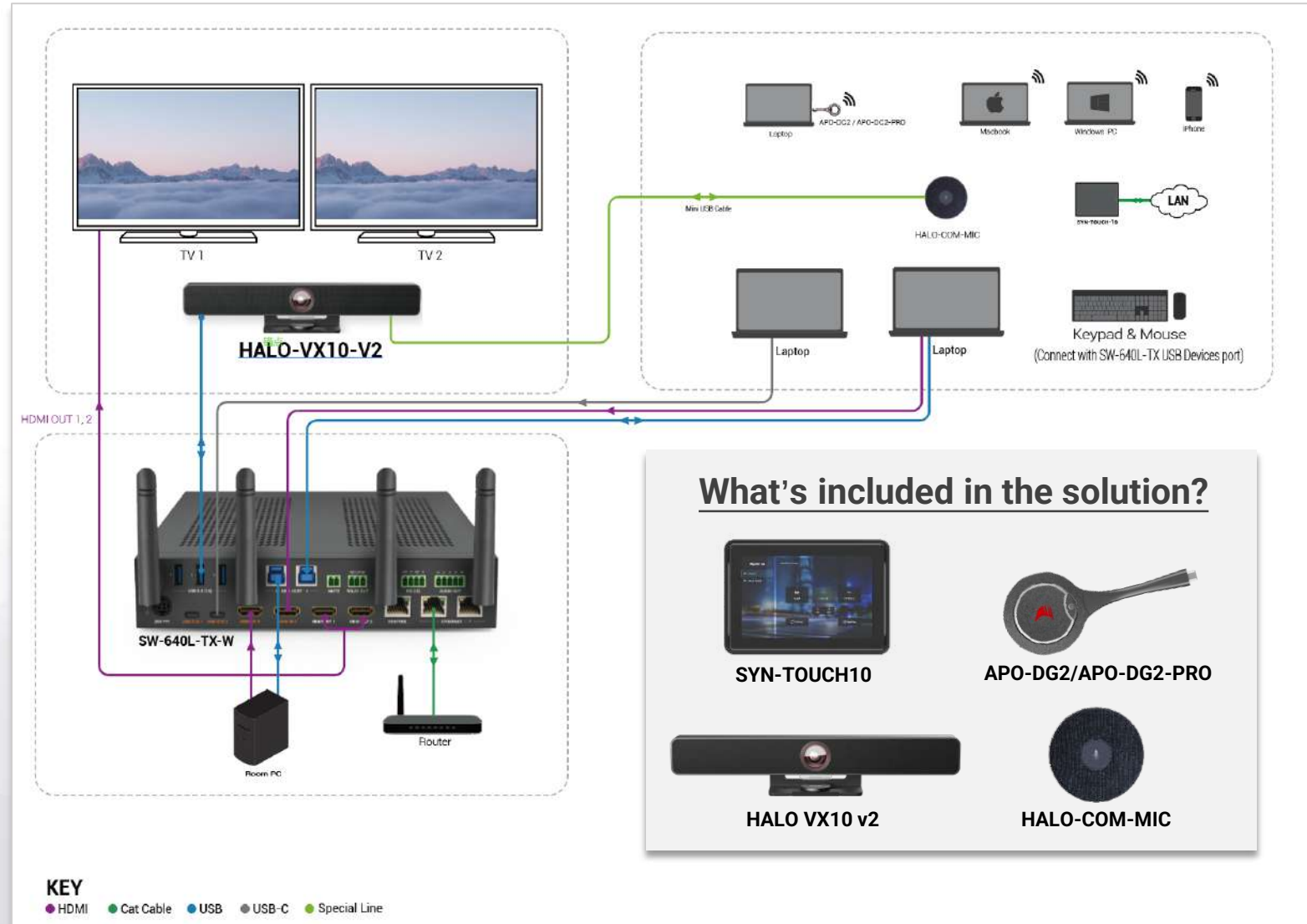
Typical Connection & Applications

Application Scenario 1: Medium Conference Room

This solution simplifies installation with its plug-and-play design, significantly enhancing meeting efficiency for both in-person and remote conferencing, ensuring every meeting is productive and smooth.

Key Benefits Include:

- **Multiple Casting Methods:** Simplifies presentations with various casting options.
- **All-in-One Video Bar:** Provides superior audio and video quality.
- **Smart Panel:** One-click control for all functions.
- **Multiview Function:** Synchronous display of multiple information sources.
- **Dual-Screen Connection:** Ensures clear and comprehensive display.
- **Fast Seamless Switching:** Quick and smooth transitions between inputs.



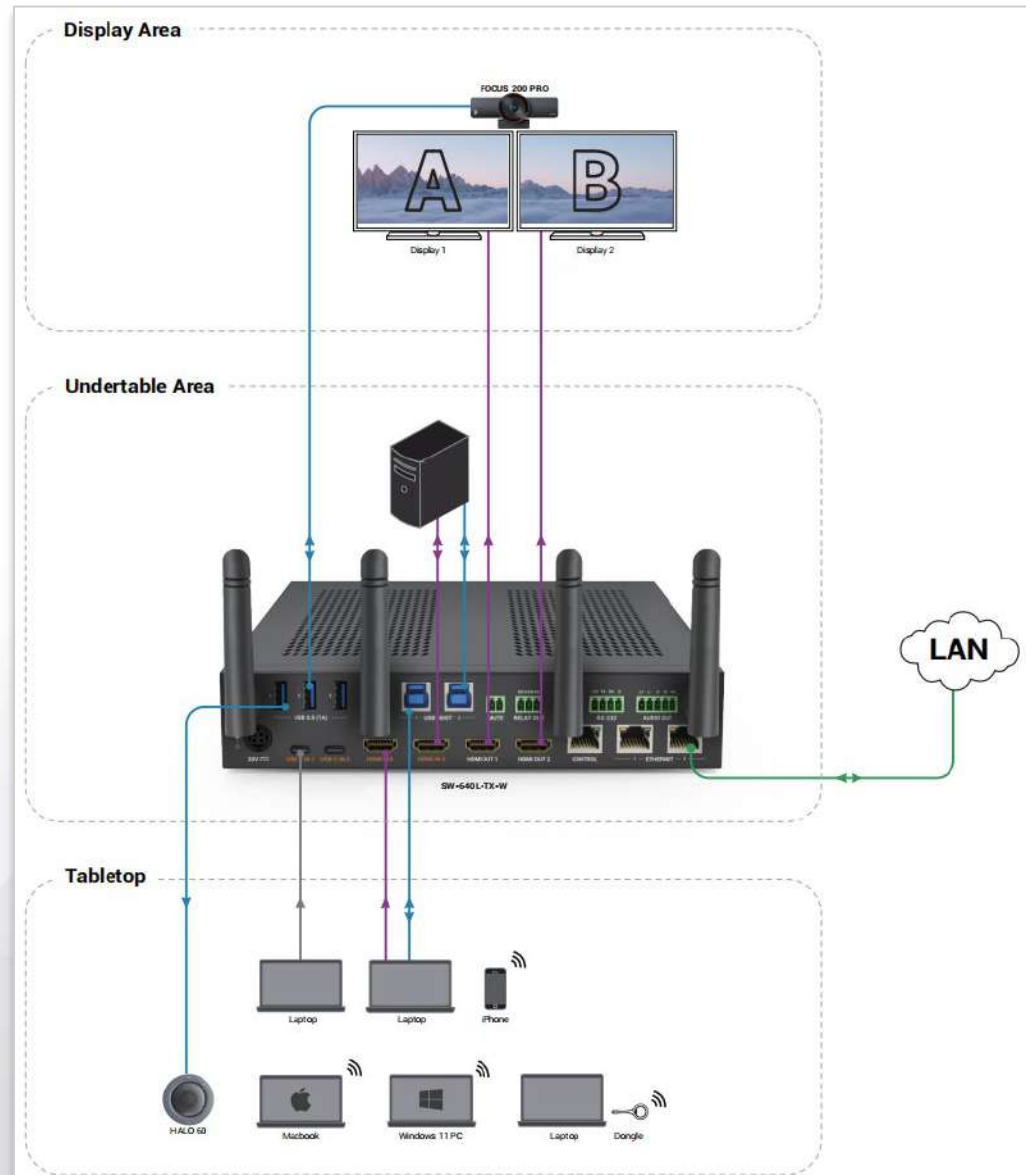
Typical Connection & Applications

Application Scenario 2: Small Classroom

This solution empowers teachers to effortlessly and quickly showcase teaching content while enabling students to interact using their own devices, enhancing overall classroom efficiency and engagement.

Key Benefits Include:

- **Multiple Casting Methods:** Simplifies presentations with various casting options.
- **Dual-Screen Connection:** Ensures clear and comprehensive display.
- **Multiview Function:** Synchronous display of multiple information sources.
- **Fast Seamless Switching:** Quick and smooth transitions between inputs.



What's included in the solution?



APO-DG2 / APO-DG2-PRO



FOCUS 200 PRO



HALO 60

History of Previous Products



SW-540-TX-W

- **Flexible Source Switching:** Automatically or manually switch between 4 connected sources (VGA, 2 HDMI, and USB-C) with assignable USB host connection.
- **Wireless Screen Sharing:** Share screens effortlessly via Apple AirPlay or Miracast.
- **Dual Output Options:** Enjoy dual output through HDBaseT and HDMI with an additional mirrored HDMI.
- **Built-in USB 2.0 Hub:** Supports USB over HDBaseT for easy connection of peripheral devices like speakerphones or cameras.
- **CEC & RS232 Triggering:** Enables automatic or manual control via CEC and RS232 triggering.



SW-220-TX-W

- **Flexible Source Switching:** Automatically or manually switch between 2 connected sources (1 HDMI & 1 USB-C).
- **Wireless Screen Sharing:** Effortlessly share screens via Apple AirPlay or Miracast.
- **Simultaneous Display:** Display up to 2 sources simultaneously, combining wired and wireless devices.
- **Built-in USB 2.0 Hub:** Supports USB over HDBaseT for easy connection of peripheral devices like speakerphones or cameras.

Comparison with Other Competitors

	WyreStorm SW-640L-TX-W	Barco CX-50 2 Gen	Creston AM-3200-WF	Barco CX-50	Extron Share Link Pro 1100
Video Input	2x USB-C (Video, USB, Ethernet, Charging) & 2x HDMI	1x USB-C (Video only)	1x HDMI	1x HDMI	1x HDMI
Input Timing	4K@30Hz	4K@30Hz	1080P@60Hz	1080P@30Hz	4K@30Hz
Video Output	2x HDMI	1x HDMI or 1x USB-C	1x HDMI	1x HDMI	1x HDMI
Output Timing	4K@60Hz (HDMI OUT1)	4K@60Hz	4K@60Hz	4K@30Hz	4K@30Hz
Wireless Casting by Dongle	Plug & Play, software-free	Non plug & play, software-dependent	Plug & play, software-free	Non plug & play, software-dependent	Unsupported
Dongle Casting Timing	4K@30Hz	4K@30Hz	4K@30Hz	1080P@60Hz	N/A
Wireless Conference	Supported, software-dependent	Supported, software-dependent	Supported, software-free	Supported, software-dependent	Unsupported
KVM Host	2x USB 3.0 Type-B 2x USB 3.0 Type-C	Unsupported	1x USB 3.0 Type-B	Unsupported	Unsupported
KVM Device	3x USB 3.0 Type-A	2x USB 3.0 Type-A 1x USB 3.0 Type-C	1x USB 3.0 Type-A 1x USB 2.0 Type-A	3x USB 2.0 Type-A 1x USB 2.0 Type-C	1x USB 3.0 Type-A 1x USB 2.0 Type-A
Host Switching	Yes	No	Yes	No	No
Network Access	3x RJ45 2x Built-in USB-C input Supports port-based physical network isolation, high security.	1x RJ45, 2.5G	2x RJ45 Guest-only network	1x RJ45	2x RJ45 Guest-only network
Multiview	Quad View	Dual View	Quad View	Dual View	Quad View
RS232	Yes	No	Yes	No	Yes
Relay	Yes	No	No	No	No
Panel Control Button	Yes	No	Yes	No	Yes

Our Advantages

This versatile system can meet all signal transmission demands in conference or classroom settings, ensuring efficient, high-quality performance for all your presentation and conferencing needs.

Our key advantages include:

- **2x Full-Featured USB-C Ports:** Provides audio/video support, USB data transfer, 60W power supply, and 1G network connections via a single USB-C cable, accommodating up to 2 laptops.
- **Multiview Support:** Display up to 4 sources on a single TV with different layout options for diverse display needs.
- **3x Ethernet Ports:** Ensures secure and stable network access for guests using their PCs, while allowing easy configuration of the system via a web interface.
- **2x HDMI Outputs:** Provides flexible output options.
- **Cost Effective:** All-in-one system at a competitive price.



Quick Installation Guide

To install the presentation device, it simply takes 4 steps below:

- **Connect and Plug in:** Follow the wiring instructions in the presentation device's quick start guide: connect the device's power interface, in-room PC, display and other conferencing devices (e.g. USB camera, speakerphone, video bar), then plug in the power adapter to turn on the device.
- **Wired Connection:** If you prefer wired connection, use USB-C or HDMI+USB cables to connect your laptop to the presentation device, then your laptop contents will be shown on the display screen.
- **Pair the Dongle:** Plug the dongle into the presentation device's front pairing port, then they will get paired automatically.
- **Wireless Casting and Conferencing:** Plug the dongle into your own laptop, then click the "casting" button to share your screen, and click the "conferencing" button to control USB peripherals wirelessly.



Quick Installation Guide

Important Notes for Installation

1. The APO-DG2-PRO dongle doesn't require any software or driver for wireless conferencing.
2. The APO-DG2 dongle requires software installation on your laptop for wireless conferencing.

For Windows Laptops:

- Insert the dongle. The system will automatically prompt you to install the software.
- Click "Confirm" to begin the installation, which will run in the background until complete.
- If no prompt appears, navigate to the "Devices and drives" section in "My Computer" to locate and install the software package.

For MacBook:

- Insert the dongle. An "Apollo Cast" disk will appear on your desktop.
 - Open the disk and follow the instructions to install the software package.
3. For wireless casting via AirPlay and Miracast, please refer to the product's user manual.
 4. For more advanced settings and configuration, you can log onto the presentation device's web interface.



Introduction of Web Configuration Interface

The Web UI of the SW-640L-TX-W presentation system is designed for intuitive management and control of the device through a web browser.

The settings and configurations you can perform are as follows:

Web UI Functionality:

- **Interface:** Manage device settings via web browsers.
- **Customization:** Personalize splash screen and network security.
- **Control:** Configure video inputs, outputs, and wireless settings.

Key Web UI Configurations:

- **General Settings:** IP configuration, security, and firmware updates, like modifying the device name, or the device's Wi-Fi hotspot name, setting the access code, etc.
- **Video Inputs/Outputs:** Manage HDMI sources and EDID profiles.
- **Screen Sharing:** Enable wireless display or multiview features.
- **Display Control:** Use CEC and RS-232 for display management, customize the guide screen picture, etc.



Final Thoughts

Q & A Session

Got questions? Let's dive into a lively Q&A session!
Your curiosity fuels the conversation, so don't hesitate to ask away.

